Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CNN/DAILYMAIL* | ***Context:*** *Ecuador Humanitarian Crisis Tweet Analysis — NGO/IO Analyst Deployment*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Inputs are formal English news articles averaging 763 tokens with 27 entities per document [Q76], processed via tokenization into Deep LSTM encoders [Q40, Q42, Q43] with entity anonymization [Q18] that suppresses lexical surface form. Empirical sampling confirms uniformly formal journalistic English register with complete sentences and standard punctuation [DATASET-D14, D15, D18]. The deployment requires processing informal-to-semi-formal Ecuadorian Spanish tweets averaging ~280 characters, with non-standard orthography, missing diacritics, crisis abbreviations, and Kichwa entity tokens (per elicitation A4 and YAML). The HF metadata explicitly tags the dataset as `language:en` and `multilinguality:monolingual` (DATASET concern 1), confirming structural absence of Spanish content. The anonymization procedure [Q18] would explicitly suppress the dialect-specific entity signals critical for Ecuadorian anchoring. This is a categorical HIGH-priority form mismatch on language, register, length, and signal type.

ID	Evidence
Q18	“To prevent such degenerate solutions and create a focused task we anonymise and randomise our corpora with the following procedure, a) use a coreference system to establish coreferents in each data point; b) replace all entities with abstract entity markers according to coreference; c) randomly permute these entity markers whenever a data point is loaded.” (p.3)
Q40	“We feed our documents one word at a time into a Deep LSTM encoder, after a delimiter we then also feed the query into the encoder. Alternatively we also experiment with processing the query then the document.” (p.4)
Q42	“We employ a Deep LSTM cell with skip connections from each input x(t) to every hidden layer, and from every hidden layer to the output y(t):” (p.5)
Q43	“"Thus our Deep LSTM Reader is defined by g^LSTM(d, q) = y(\” (p.5)
Q76	“Note that these examples were chosen as they were short, the average CNN validation document contained 763 tokens and 27 entities, thus most instances were significantly harder to answer than these examples.” (p.8)
D14	The country is at a critical and dangerous juncture -- threatened by rising tensions and spreading terrorism, Musharraf said in a televised address to the nation after declaring martial law. — Political crisis description in formal English news register — contrasts with informal Ecuadorian Spanish tweet register
D15	Miami Police Officer Gil Gonzalez said Lewis 'had a Bible and was praying when we went to get him. He had a look of guilt, a look of shock.' — Formal English journalistic prose; illustrates register mismatch with informal crisis tweets
D18	There was no 'deliberate violence,' committed by Blackwater employees, he added. Still, when asked whether it is possible someone with Blackwater 'screwed up' in the incident, Prince replied, 'Certainly it's possible.' — Direct quotation in US English journalism; illustrates that all content is English formal register

Strengths

- The benchmark documents its sensitivity to document length and entity density [Q76, Q78], providing methodological transparency about input characteristics — useful as a contrastive anchor when evaluating tweet-length systems.

ID	Evidence
Q76	“Note that these examples were chosen as they were short, the average CNN validation document contained 763 tokens and 27 entities, thus most instances were significantly harder to answer than these examples.” (p.8)

Q78	“To understand how the model performance depends on the size of the context, we plot performance versus document lengths in Figures 4 and 5. The first figure (Fig. 4) plots a sliding window of performance across document length, showing that performance of the attentive models degrades slightly as documents increase in length. The second figure (Fig. 5) shows the cumulative performance with documents up to length N, showing that while the length does impact the models' performance, that effect becomes negligible after reaching a length of ~500 tokens.” (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Severe mismatch — source is formal English news prose at ~763 tokens [Q76]; target is informal Ecuadorian Spanish tweets at ~40–60 tokens with non-standard orthography, missing diacritics, and crisis abbreviations [WEB-12, elicitation A4].

IF-2: Regional infrastructure (Twitter/X access, ~78.2% internet penetration [WEB-12], ~57% smartphone penetration [WEB-14]) supports tweet capture, but the benchmark's signal specification (formal long-form English news) is not what regional infrastructure produces during crises.

IF-3: Domain-specific differences: (a) language mismatch (English vs Spanish/Kichwa-influenced); (b) register mismatch (formal news vs informal crisis-time social media); (c) length mismatch (763 tokens vs ~50 tokens); (d) entity-anonymization procedure [Q18] would suppress diagnostic dialect-specific signals; (e) crisis-time connectivity disruptions create survivorship bias [WEB-12] absent from news data.

IF-4: All four form mismatches (language, register, length, anonymization-induced signal suppression) constitute severe external validity violations for the deployment.

ID	Evidence
Q18	“To prevent such degenerate solutions and create a focused task we anonymise and randomise our corpora with the following procedure, a) use a coreference system to establish coreferents in each data point; b) replace all entities with abstract entity markers according to coreference; c) randomly permute these entity markers whenever a data point is loaded.” (p.3)
Q56	“The entity anonymisation and permutation aspect of the task presented here may end up levelling the playing field in that regard, favouring models capable of dealing with syntax rather than just semantics.” (p.6)
Q76	“Note that these examples were chosen as they were short, the average CNN validation document contained 763 tokens and 27 entities, thus most instances were significantly harder to answer than these examples.” (p.8)
WEB-8	No NLP resources existed for Kichwa before 2024 Killkan ASR dataset (https://arxiv.org/abs/2404.15501)
WEB-10	Kichwa agglutinative morphology creates context-dependent forms that compound NLP challenges (https://cdt.org/wp-content/uploads/2025/06/2025-06-25-Quechua-Report-English-final.pdf)
WEB-12	Internet penetration 78.2%, with crisis-time power outages disrupting telecom (https://freedomhouse.org/country/ecuador/freedom-net/2024)
WEB-14	Smartphone penetration ~57% nationally (https://www.statista.com/statistics/1081753/smartphone-owners-ecuador/)

Information Gaps

- Specific orthographic-degradation patterns in Ecuadorian crisis-time tweets (deferred per YAML)

Requires Expert Verification

- Field-analyst confirmation of Kichwa-Spanish code-switching patterns observed in actual SGR-tracked event tweet streams

Remediation

Gap 1: Formal English news at ~763 tokens with entity anonymization; no Spanish, no informal register, no tweet-length inputs

Recommendation: Replace the input form entirely with informal Ecuadorian Spanish tweets at native length distribution, preserving non-standard orthography, missing diacritics, crisis abbreviations (SGR, COE, EDAN, GAD, MSP), and Kichwa-Spanish lexical mixing; do not apply entity anonymization, since dialect-specific surface forms are diagnostically critical.